Augmented Reality Visualization of Menu Items using Image Reconstruction

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ABSTRACT

In today's fast-paced world, the abundance of dining options often overwhelms customers, making decision-making challenging. Traditional menus have 2D images that lack the depth and immersion needed for an accurate visual representation of each dish. Our ground-breaking approach utilizes state-of-the-art technology, converting restaurant-uploaded 2D images into 3D models. Through our mobile application and a QR code on the table, customers can visualize dishes, empowering them to make informed decisions before ordering. As a result, this application which is developed using Unity 3D, AliceVision and AR frameworks, transforms the dining experience. Our technology improves overall customer pleasure and engagement with the eating process by providing customers with comprehensive insights on meals, therefore transforming the way customers interact with restaurant menus.

CCS CONCEPTS

• **Human-centered computing** → Mixed / augmented reality; • **Computing methodologies** → Reconstruction.

KEYWORDS

Augmented Reality, Visualization, Unity3D, 3D Image Reconstruction, AliceVision, Photogrammetry

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1 INTRODUCTION

In today's rapidly evolving technological landscape, Augmented Reality (AR) has emerged as a powerful tool with diverse applications across various industries. AR creates new levels of interaction and improves user experiences by smoothly combining digital and physical information. Its developing prevalence in day-to-day life is apparent in a variety of industries, including gaming, education, healthcare, and retail.

AR's capacity to provide immersive and interactive experiences is one of the main factors contributing to its growing popularity. Applications nowadays range from augmented reality games like Pokémon GO to learning resources that let users concretely see difficult ideas. AR has also made its way into the retail industry, allowing virtual try-ons for apparel and accessories, completely changing the way buyers make decisions.

AR is a rising star in the food industry, changing how consumers interact with menus and culinary options. Restaurants can overcome language barriers and uncertainty associated with traditional 2D menus by utilising AR technology to provide customers with interactive and realistic visualisations of menu items. This makes the process of making decisions easier and also makes the dining experience more enjoyable and memorable.

Within the framework of the constantly changing dining experience, one recurring issue is when customers are taken by surprise by the size or presentation of the food they have ordered, or find it difficult to understand the menu due to language barriers. All of these problems add up to a big difference in the way users experience ordering food. Our creative idea suggests a way to solve this, which is to visualise the food before placing an order.

Utilising 3D image reconstruction technology, a novel technique meant to produce an immersive and lifelike representation of menu items is the foundation of our approach. We hope to overcome the drawbacks of conventional 2D menus and enable users to make better decisions by utilising this cutting-edge technology. By enabling customers to view meals in three dimensions and get a realistic sense of portion sizes and presentation, our AR solution goes beyond traditional menus. This tactical change not only improves decision-making but also helps to reduce overordering tendencies, which makes meal selection more effective and sustainable while reducing food waste.

With our methodology, restaurant owners submit their 2D menu photos in an easy-to-use and efficient manner. The 3D reconstruction engine carefully examines these photos, taking out important details to produce accurate and realistic 3D models of every menu item. By utilising the Unity platform, our augmented reality models allow users to engage with three-dimensional representations of meals in a physical environment, thereby revolutionising the dining experience.

This study demonstrates how augmented reality menus with 3D image reconstruction can revolutionise the way customers interact with restaurant offerings. Our ultimate goal is to integrate cutting-edge 3D reconstruction technology to offer restaurant owners an augmented reality experience that is more engaging and intuitive. By doing this, we hope to close the gap that currently exists between traditional 2D menus and the changing needs of modern consumers, revolutionising dining and raising the bar for culinary engagement.

2 RELATED WORK

In the domain of 3D model reconstruction from 2D images, [1] proposes a non-contact, non-invasive method emphasizing image quality. Despite its effectiveness in generating accurate 3D models, challenges such as computational demands and potential hardware/software requirements underscore the intricacies of the approach. The approach also uses visual hull algorithms that lead to excessive volume approximation due to object concavity and insufficient perspective views, introducing inaccuracies. Additionally, their deterministic nature overlooks uncertainties in views and makes them highly sensitive to segmentation errors, where a single error can significantly impact the generated 3D model. Evaluation metrics, Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) comprehensively assess its performance. Acknowledging limitations, the method is transparent about its dependence on both image quality/quantity and computational complexity.

A Mesh RCNN which is a 3D shape prediction machine learning (ML) model facilitates object recognition and categorization in the system [2]. Utilizing a CNN on the 2D image, depth is determined, and objects in the scene are distinguished. The 3D model is then constructed based on this 2D image, recreating the scene using voxels to address obscured areas. The authors evaluated PyTorch3D against Open3D, MeshCNN (a neural network architecture for processing and learning from 3D mesh data), and PyTorch Geometric for various 3D deep learning tasks. Evaluation criteria included accuracy, speed, and memory utilization. PyTorch3D outperforms other 3D deep learning libraries in precision, speed, and memory consumption based on experimental data. The authors suggest its potential application in industries such as computer vision, robotics, and graphics.

As a survey of various 3D reconstruction pipelines, [3] presents a comparison of Autosweep [8], Point Cloud [9], Pixel2Mesh [10], Mesh-RCNN [11] and Pytorch3D [2] methods. The study states the supremacy of Pytorch3D methods by thorough evaluation and substantial proof. This paper supports the claims in [2] on a more diverse dataset. Pytorch3D's automated methods involving noise removal, feature enhancement, Scale-Invariant Feature Transform (SIFT) or Speeded Up Robust Features (SURF) based feature extraction, k-means clustering, 3D coordinate estimation, Levenberg-Marquardt optimization, and smooth 3D model generation, excel in complex scenes with applications in architecture, engineering, and entertainment. The method proves superior in accuracy and effectiveness compared to existing 3D reconstruction techniques. The authors envision broad applications across industries such as engineering, entertainment, and architecture, highlighting its potential impact.

Exploring AR entails researching the most recent programs, frameworks, and machine-learning approaches for critical Mobile AR components [4]. The research plan proposed in the paper is for comprehensive MAR frameworks and ML-assisted pipelines for adaptive AR experiences. The authors envision MAR transforming how people interact with the physical and digital worlds, but they emphasise that it must overcome technological and usability problems before it can be fully realised. The report offers further research in MAR, with an emphasis on improving user experience, intelligence integration, and tracking accuracy.

The use of augmented reality (AR) to improve smooth navigation in enclosed spaces is highlighted in [5], an exploration of the use of AR for indoor navigation. The research looks at techniques and algorithms for AR-based indoor navigation, showing practical applications and emphasising AR's promise for real-time navigation help in interior areas. Overall, the research makes a substantial contribution to the field of augmented reality applications, notably in terms of revolutionising indoor navigation and encouraging additional research in this growing area.

The issues of virtual navigation and object interaction are addressed in [6] by introducing a navigation framework for Virtual Reality (VR), an application of extended reality. The framework aims to improve the VR user experience by utilising cutting-edge technology and offers creative methods for smooth navigation and effective interaction with virtual items. The paper provides both theoretical insights and practical implementations, paving the way for advancements in VR, particularly in navigation and object interaction domains.

3 MATERIALS AND METHODS

This section explores the problem statement, in-depth architectural insights, the working of the reconstruction engine, the front end of the application and the menu database.

The methodology for implementing an AR food menu with 3D image reconstruction involves restaurant owners submitting high-quality 2D images of menu items. A 3D reconstruction engine analyzes these images, creating accurate 3D models. These models are then integrated into an AR environment using the Unity platform, allowing customers to interact with lifelike representations of menu items through a dedicated mobile application or QR code scanning. Thorough testing, user education, and continuous monitoring ensure a seamless and engaging AR dining experience, ultimately bridging the gap between traditional 2D menus and the evolving expectations of contemporary consumers.

The creation of AR food menu system can be broadly classified into the creation of the 3D reconstruction engine that uses 2D images of the menu item and converts it into a 3D model; the unity front-end for the AR visualization of the menu items and a server for communication. To ensure a varied depiction of food, we intend to gather 2D photos of menu items from restaurant owners. These pictures will be the source data used by our in-house 3D image reconstruction engine.

For the creation of a robust 3D reconstruction engine capable of transforming 2D menu images into realistic 3D models, we have utilized the framework built by AliceVision. Meshroom [7] applies advanced computer vision techniques to extract key attributes and dimensions from these photographs to accurately portray menu items. The framework integrates diverse machine-learning techniques within a photogrammetry pipeline implemented in C++. It leverages the capabilities of OpenCV (an Open-Source Computer Vision Library) and harnesses GPU-accelerated CUDA (Compute Unified Device Architecture, a parallel computing platform created by NVIDIA) programs to efficiently create 3D models. Meshroom's nodal architecture allows us to customize the reconstruction pipeline to adjust it to our domain-specific needs. AliceVision has been conceived as a complete open-source photogrammetry framework that bridges the gap between academic research and professional demands. Each step of the reconstruction pipeline has a default set of optimised algorithms for improved performance and quality of the resultant 3D models. Thus we have incorporated Meshroom's open-source pipeline in bringing our application to life.

The front-end is built with Unity3D. Unity3D, commonly referred to as Unity, is a popular cross-platform game development engine that is widely used to create 2D and 3D games, simulations, and interactive experiences (AR/VR). This is where our 3D reconstructions come to life. Unity will be the basis for building augmented reality models that blend in with the actual dining setting. This front-end is constructed with meticulous attention to detail, ensuring a user-friendly and interactive interface that makes it convenient for customers to engage with 3D representations of menu items.

A server is built using Python's Flask (a lightweight web framework that is designed to make it easy to develop web applications) to bridge the communication between the 3D reconstruction engine and the front-end.

Comprehensive testing and optimisation are essential components of our work. To ensure the accuracy and realism of our 3D reconstructions we have tuned various parameters. To improve the User Interface (UI) and overall experience user-level tests were made. The main goals of optimisation will be responsiveness, speed, and interoperability with different AR-capable devices.

Furthermore, we recognize the significance of cloud storage in our system. Scalability and accessibility will be guaranteed by the safe cloud environment in which the reconstructed models will be kept. This cloud-based method makes it possible to retrieve 3D models quickly and easily in various devices, which increases the overall adaptability of our augmented reality menu system.

Creating 3D models with software like Blender or Maya demands human expertise and time-consuming efforts. The process can be both intricate and expensive due to the need for skilled labour. Thus, In summary, our innovation revolves around automating 3D model generation for practical application in the food industry. Driven by addressing customer needs, we've developed a user-friendly platform tailored for the convenience of both restaurant owners and customers.

3.1 System Architecture

This section describes the system's architectural diagram and flow. In Figure 1, the interaction between two user types is illustrated: Customers, who are individuals visiting a restaurant, and Restaurant Owners, who own a restaurant. Both user types engage with the user interface, which is linked to the backend database and reconstruction engine through a communication server.

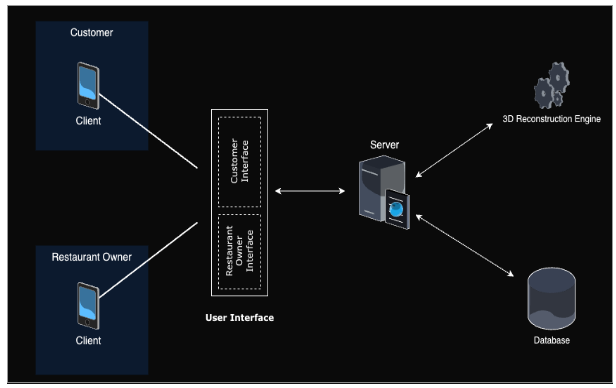

Figure 1: Architectural diagram

There are two main parts: the 3D image reconstruction engine and the UI that is created using Unity3D.

UI for AR Rendering - A web application to facilitate the seamless submission of 2D images by restaurant owners. Users will have the convenient option to browse and select desired items from a user-friendly menu available on the website along with dish information.

3.2 Reconstruction Engine

The 3D reconstruction engine will harness machine learning capabilities to craft a model that can analyze diverse images, identifying dishes and meticulously mapping vertices and textures to construct intricate 3D models. To ensure a cohesive workflow, a Python web framework will serve as the backbone, seamlessly connecting each step. This not only facilitates a user-friendly interface but also ensures efficient data processing, laying the groundwork for a streamlined and effective system.

The following sections will provide concise descriptions of the various steps in the reconstruction process shown in Figure 2. Each step employs distinct algorithms, contributing incrementally to the overall construction of the model. The output of each step serves as the input for the subsequent one in a streamlined and interconnected fashion.

3.2.1 Feature Extraction. Identifying distinguishing points or features in 2D images is the first step in the image reconstruction process. To robustly identify and describe important points in the images, Alice Vision uses sophisticated feature extraction algorithms like SIFT or SURF. We have included the pseudo-code for SIFT.

```
1. Start.
```

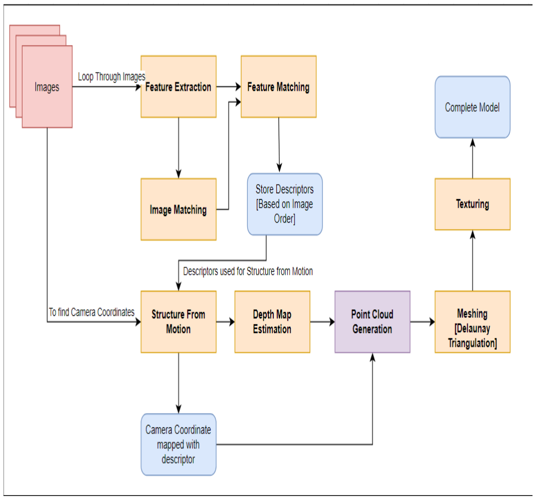

Figure 2: Reconstruction Process

```
2. Check if the number of images is more than 40.
3. If yes, group images into batches of size 40.
4. For every image in the batch:
a. Convert the image to the SRGB format.
b. Compute and store the number of octaves.
c. Generate Gaussian images using octaves.
5. For each adjacent pair of images in the batch:
a. Subtract and store adjacent pairs as DoG (Difference of Gaussian).
6. Store keypoints using Gaussian images, DoG, and the number of octaves.
7. Remove duplicate keypoints.
8. Resize keypoints to the image size.
9. Store descriptors using keypoints and Gaussian images.
10. Write keypoints and descriptors in sift.desc.
11. Stop.
```

3.2.2 Camera Pose Estimation. The next stage is to estimate the camera poses using the features that have been extracted, which entails figuring out the location and orientation of the cameras that took the pictures. Even in the face of outliers or noisy data, Alice Vision can estimate precise camera poses thanks to strong algorithms like RANSAC (Random Sample Consensus).

3.2.3 Structure from Motion (SfM). Alice Vision uses SfM methods after estimating the camera poses. SfM triangulates the matching 2D points in several images to reconstruct the scene's 3D structure. Iteratively, it improves the 3D structure and camera angles to produce a realistic portrayal of the scene. Pseudo-Code for Sfm is given.

```
1. Start.
2. Get sift.desc, feature_matches.txt, and images.
3. For every feature match in feature_match.txt:
a. Estimate the Essential Matrix using corresponding descriptors.
b. Estimate the Fundamental Matrix using descriptors in image matches.
c. Estimate the world pose from 3D to 2D point correspondences.
d. Estimate the relative transformation between camera poses.
e. Push the relative transformation into POINT_TRACK.
4. Call Triangulate_Multi-View.
5. Call bundleAdjustment, which adjusts the collection of 3D points.
6. Stop.
```

3.2.4 Depth Map Estimation. An essential first step in comprehending the spatial relationships within the reconstructed scene is depth map estimation. Alice Vision determines the distance information for each pixel in the images using depth map estimation algorithms, potentially with the help of stereo-matching techniques. This data helps to build a more intricate 3D model by capturing the relative distances between objects. The accuracy of the depth map significantly influences the realism and precision of the reconstructed scene, impacting subsequent stages of the reconstruction pipeline.

3.2.5 Meshing. The creation of a seamless and uninterrupted 3D mesh from the reconstructed points, known as meshing, is a pivotal step in the process. Alice Vision employs sophisticated algorithms to construct a mesh that accurately represents the surfaces and contours of the reconstructed scene. This crucial phase is essential for crafting a 3D model that not only adheres to geometric precision but also ensures aesthetic appeal, underscoring its significance in the overall modelling process.

3.3 Unity front-end

The Unity front-end plays a key role in our application, bridging the gap between the physical and virtual worlds to offer users an immersive AR dining experience. The key actions and functionalities integrated into our Unity front-end are described, with special attention to plane detection, dynamic button creation, database integration and user interaction with 3D models. These elements collectively contribute to a seamless and interactive interface, enhancing overall user engagement and ensuring a compelling AR experience for exploring and interacting with menu items.

3.3.1 AR Foundation Integration. AR Foundation is a cross-platform AR framework developed by Unity Technologies. It allows developers to create AR experiences that can run on both iOS (ARKit) and Android (ARCore) devices using a single codebase in Unity. With the integration of AR Foundation, our Unity front end can now support augmented reality on a variety of devices. This fundamental stage establishes the foundation for a seamless user experience by enabling features like AR interactions and plane detection.

3.3.2 AR Plane Detection. Our application uses AR plane detection to find flat surfaces in the user's surroundings, adding to the realism. With the help of this feature, we can bind virtual menu items to actual surfaces to create a tangible and engaging experience. The pseudo-code shows how we utilize AR Frameworks for

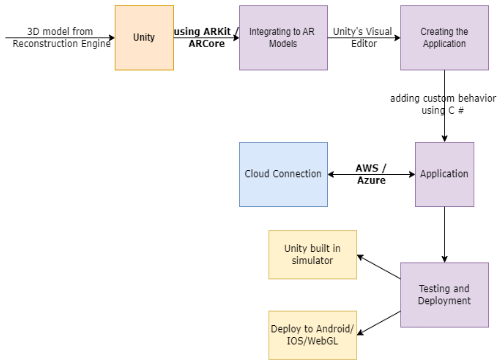

Figure 3: Unity Overview

detecting a plane surface(such as a restaurant table) and placing the 3D-spawned object.

```
1. Start
a. Declare GameObjects: arObjectToSpawn, placementIndicator, spawnedObject
b. Declare Pose: PlacementPose
c. Declare ARRaycastManager: aRRaycastManager
d. Declare bool: placementPoseIsValid=false
2. Function Start(): Initialize aRRaycastManager with an instance of ARRaycastManager found in the scene
3. Function Update():
a. If spawnedObject is null, placementPoseIsValid is true, and there is at least one touch with Began phase:
b. Call ARPlaceObject()
c. Call UpdatePlacementPose()
d. Call UpdatePlacementIndicator()
4. Function UpdatePlacementIndicator():
a. If spawnedObject is null and placementPoseIsValid is true:
b. Activate placementIndicator, set its position, and rotation to PlacementPose
c. Else:
d. Deactivate placementIndicator
5. Function UpdatePlacementPose():
a. Calculate screenCenter as the screen point corresponding to the camera's center
b. Create a list of ARRayManager at screenCenter onto detected planes
c. Set placementPoseIsValid to true if the count of hits is greater than 0
d. If placementPoseIsValid is true, set PlacementPose to the pose of the first hit
6. Function ARPlaceObject():
a. Instantiate arObjectToSpawn at PlacementPose's position and rotation, and assign the instance to spawnedObject
7. End
```

3.3.3 Dynamic Button Generation. In each restaurant, items are organized into various categories. When a customer scans a QR code on a restaurant table, they should have access to the restaurant's menu, neatly sorted by categories. Similarly, for a restaurant owner logging in, the menu items should be presented in a categorized manner. To achieve user-friendly navigation, buttons are dynamically generated within the Unity scene. These buttons correspond to the items extracted from the database, making them clickable. Each button is instantiated with labels and textures that match the respective menu items, ensuring an intuitive and seamless experience for users exploring the menu.

3.3.4 Database Connection. In our application, the Unity front end connects to the database using UnityWebRequest, enabling the dynamic retrieval of essential information about 3D models, such as menu item textures and coordinates. Communication between Unity and the backend is facilitated by a Flask server, with data exchanged in JSON format. The Unity front end triggers HTTP requests to the Flask server (GET or POST), which processes these requests, interacts with the database, and returns relevant information in JSON. This seamless interaction ensures real-time updates, enhancing the user experience in the 3D environment.

3.3.5 Interaction with 3D Models. Unity front-end utilizes information from the database to effortlessly load 3D models into the scene. This dynamic instantiation enables users to explore and interact with realistic representations of menu items in their augmented reality environment. To ensure spatial accuracy, our application anchors 3D models to the detected AR plane. This spatial anchoring allows virtual objects to seamlessly integrate into the physical world, providing stability and enhancing the user's perception of virtual elements.

In crafting the user interaction, ease of use is prioritized. Button placement, labelling, and visual feedback were thoughtfully considered in the AR menu design, contributing to an optimized and user-friendly experience. The intuitive interface ensures that users can navigate and engage with the AR menu effortlessly, enhancing overall satisfaction and usability.

3.4 Database

Our application's MySQL database acts as a location for crucial data, which improves the overall performance of our system. It effectively organises object files, which contain intricate 3D representations of the menu items connected to every restaurant. Additionally, restaurant results are stored in the database, resulting in a structured dataset that can be easily retrieved and presented. By creating and storing QR codes in the database, each restaurant is given a unique identification that provides a rapid and safe method of access. Furthermore, each restaurant's menus are carefully catalogued by the MySQL database, guaranteeing a centralised and well-organized collection of food options. The pseudo-code describes how our database is queried to get all items from a particular restaurant menu.

```
1. Start
2. Function get_restaurant_menu(db_connection)
a. TRY
i. SET cursor as db_connection.cursor(dictionary=True)
ii. SET query as "SELECT * FROM menu_items WHERE rid = %s"
iii. CALL cursor.execute(query)
iv. SET items to cursor.fetchall()
```

```
v. IF NOT items
vi. RETURN 404, DATA NOT FOUND
vii. RETURN 200, items
b. EXCEPT err
c. RETURN 400, BAD REQUEST
3. End
```

The seamless integration of augmented reality menus into the dining experience is made possible by this architecture, which not only simplifies data management but also allows for quick and accurate retrieval.

3.5 Cloud Connection

Our application uses AWS cloud service to provide cloud connectivity. This service is used to store data restaurant and menu data. We also use an S3 bucket to store 3D models. This allows to accommodate many restaurants and customers in real-time. We also host our Flask server and the reconstruction engine in a compute instance. Unity provides necessary integration support.

3.6 Data

Our data contains high-quality photographs recorded with a DSLR or other excellent camera from all angles. The photographs should encompass all portions of the dish, and the person taking them should make sure there are no shadows on the image and that the surrounding area is well-illuminated and has adequate lighting. We anticipate that restaurant owners will transmit a minimum of nine photographs in order to obtain a cleaner outcome. We expect restaurant owners to submit numerous well-lit, shadow-free, high-quality photos of dishes, captured from all angles for optimal results.

4 IMPLEMENTATION AND RESULTS

The implementation of the system encompasses the techniques and technologies elaborated in the materials and methods. As an overview, AR Food Menu comes to life utilizing Unity3D for the user interface; OpenCV and CUDA for reconstruction; and Flask as a communication server for our SQL database and reconstruction engine. The utilization of Unity3D for the user interface ensures a seamless and engaging user experience, while the integration of OpenCV and CUDA for reconstruction leverages GPU's(Graphics Processing Unit) capabilities. The choice of Flask as a communication server adds an extra layer of flexibility, enabling smooth interaction between our SQL database and the reconstruction engine. Careful design philosophy is applied with the front end using an object-oriented approach and the Reconstruction Engine using a nodal structure. By utilising the modular architecture of OpenCV, multiple interconnected modules exchange information via the Reconstruction Interface. The server and Unity application use the Model-View-Controller Pattern of Design. It separates an application into three interconnected components- Model, View and Controller to facilitate modularization, scalability and separation of concerns thus enabling us to own a maintainable codebase.

The final image generated by our image reconstruction engine illustrates the successful identification and rendering of 2D images in a real-world setting in the visual representation of our study's results (see Figure 3) This picture offers a concrete example of how efficient the system is at capturing the minute details of different items. The dish represents Pakora (Pakora is a fritter originating from the Indian subcontinent. They are sold by street vendors and served in restaurants in South Asia. It consists of items, often vegetables such as potatoes and onions, coated in seasoned gram flour batter and deep fried). Figure 5 displays the user interface provided by our application. The application includes a common landing page for both customers at a restaurant and restaurant owners. Here, customers can choose to scan the QR code placed on their table, enabling them to view a list of menu items. Restaurant owners can log in, view their menu, or add a new menu item by filling in the details and uploading images of the dish. The immersive and interactive qualities of our AR food menu system are emphasised through the app's UI.

Furthermore, Figures 6 and 7 show our AR app in use, the smooth integration of virtual food items into the real world not only adds a layer of excitement to the dining experience but also underscores the practical applications of augmented reality in the culinary domain. These pictures provide strong visual proof of our system's capabilities, supporting the favourable results seen in the performance and user experience metrics.

Customers can enjoy the ease of using our QR code scanning feature, effortlessly accessing an immersive menu. On the other hand, restaurant owners find convenience in a robust set of tools, making menu customization and management easy. This technology empowers restaurants to create lasting memories for customers, improving their overall experience. Doing so, not only boosts positive customer reviews but also contributes to the growth of the business on a larger scale. Additionally, it relieves restaurant owners from the burden of managing and updating a traditional 2D menu.

4.1 Hardware and Software System Requirements

Hardware and software requirements refer to the specifications and capabilities necessary for a device to run our application. These requirements are essential to ensure optimal performance, compatibility, and functionality. Table 1 breaks down these requirements for each user category of our application. Customers visiting the restaurant would only require a smartphone with either an Android or iOS operating system whose specifics are shown in Table 1. A restaurant owner would additionally need a good camera for capturing images of the food items. A developer on the other hand will additionally need the software(s) mentioned in Table 1.

5 CONCLUSION

In conclusion, the incorporation of 3D image reconstruction and AR not only transforms the dining experience but also has enormous potential for wider applications in a variety of fields. Beyond the culinary realm, this innovative technology could redefine how users interact with a myriad of products and services. Imagine realistic AR shopping experiences, enabled by accurate 3D reconstructions, where customers can virtually try on clothes or see furniture in their homes before making a purchase. By giving students realistic 3D models for in-depth study of difficult subjects, this technology

Table 1: System Requirements

Developer	Restaurant Owner	Customer
Unity3D and C# for front-end, MySQL for database	Smart Phone	Smart Phone
C++, OpenCV, and CUDA setup for reconstruction	Android 7.0 and above, iOS 6.1 and above	Android 7.0 and above, iOS 6.1 and above
Python and its libraries for server	Good Camera (DSLR/SLR)	

Figure 4: Final output of the reconstruction engine after meshing

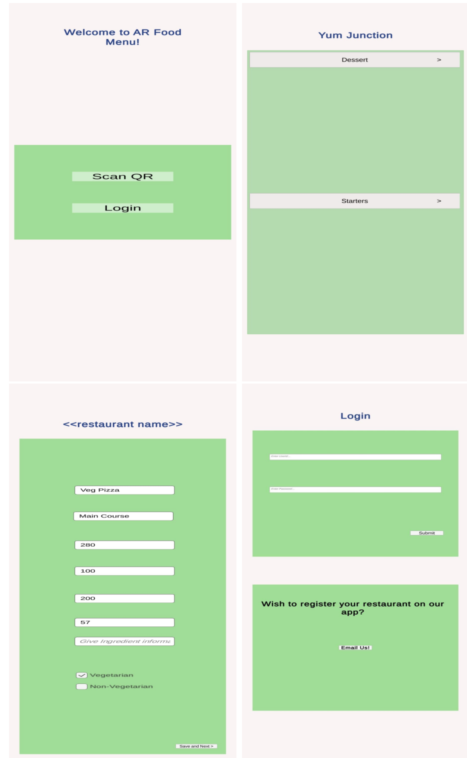

Figure 5: Application Interface

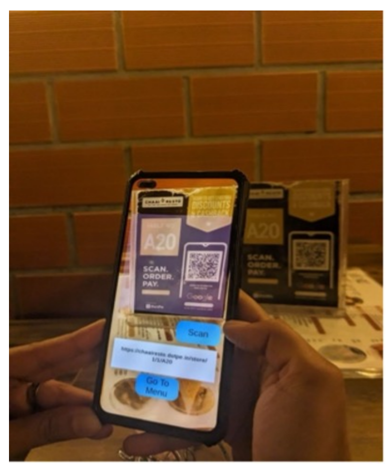

Figure 6: Image of scanning the menu

has the potential to revolutionise education by improving comprehension and engagement.

Furthermore, AR and image reconstruction in the medical field may help with realistic organ models during surgical planning, leading to more accurate medical interventions. The amalgamation of augmented reality and 3D image reconstruction unfolds into a transformative force with implications spanning across industries, promising a future where technology seamlessly integrates with our daily lives. Additionally, the synergy between augmented reality and 3D image reconstruction not only promises transformative applications in education and healthcare but also hints at potential breakthroughs in fields such as architecture, gaming, and design, further enriching our technological landscape.

Future work entails Integration of AR Food Menu with Online Food Ordering Apps- Collaborate with online food ordering apps to enable direct ordering from the AR interface; Integration with Ordering Systems- Collaborate with restaurant ordering systems to enable direct ordering from the AR interface; and Introduce accessibility features, such as voice commands and text-to-speech capabilities, to make the AR Food Menu more inclusive for users with diverse needs.

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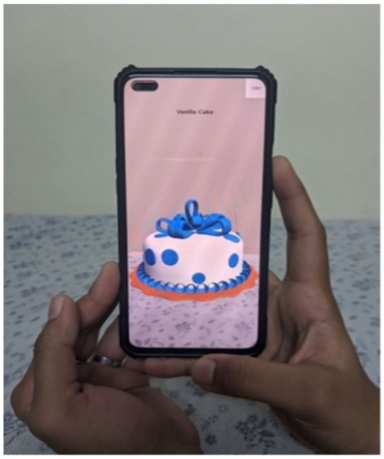

Figure 7: Image of vanilla cake model

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